ES 225 – AI and Data Science

Term Project – Assignment #2

Descriptive and Exploratory Analysis of Surface Water Quality

Kerala CWC Dataset with Ramamangalam Station Analysis

Submitted in partial fulfilment of the requirements for the course:
ES 225 – AI and Data Science

Group 10

State / Dataset: Kerala – Water Quality

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Data Availability

The full raw dataset and the cleaned, station-filtered dataset for Ramamangalam used throughout this report are available at the following Google Drive folder:

`https://drive.google.com/drive/folders/1Iy9n2XDmbHdw5mTXCZkmGpRbUS0Pes2e?usp=drive_link`

The workbook (`Ramamangalam_Water_Quality_Descriptive_Stats_CLEANED.xlsx`), contains three sheets:

1. **Ramamangalam_Raw** – all 359 Ramamangalam observations, organized chronologically where possible.
2. **Sorted_Numeric_Data** – each numerical parameter independently sorted from smallest to largest, making median, Q1, and Q3 easy for the TA to manually verify.
3. **Descriptive_Stats** – Mean, Median, Mode, Sample SD, Min, Max, Range, Sample Variance, Q1, and Q3 for each parameter.

Contribution of the Group Members

Name	Roll No.	Section
Tanmay Sandip Dhule	25B4208	1.1–1.3 Descriptive Statistics + report compilation + Google Drive Link
Teertha Pankaj Gandhi	25B4233	1.4–1.6 Outliers, Correlation Matrix, Skewness/Kurtosis
Sri Vishnu Vikyath S	25B4246	2.1–2.2 Histograms, Station-wise Boxplots
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Objectives – Assignment #2

The primary objective of this project is to perform descriptive and exploratory statistical analysis of surface-water-quality parameters using the Kerala CWC dataset, including an overall analysis of all available monitoring stations and a station-specific analysis of the Ramamangalam monitoring station.

The specific objectives, following the assignment brief, are to:

(a) Compute descriptive statistics (mean, median, mode, standard deviation, minimum, maximum, range, variance, and quartiles) for each numerical water quality parameter.

(b) Detect outliers using the IQR method or z-scores, and compute the correlation matrix to identify the strongest inter-parameter relationships.

(c) Compare skewness and kurtosis across parameters to characterize the shape of each distribution.

(d) Visualize the data through histograms, station-wise boxplots, a correlation heatmap, temporal line plots, and univariate scatter plots.

(e) Interpret the resulting hydro-chemical patterns at Ramamangalam, distinguishing plausible natural (hydrological, seasonal) drivers from anthropogenic influences, and flagging anomalous or isolated observations.

Chapter 1

Descriptive Statistics

1.1 Dataset Summary

This descriptive analysis establishes both an overall baseline for the region and a specific evaluation of the assigned station. The scope is divided into two parts:

- **Full Kerala CWC Dataset (All Stations Combined):** This analysis uses the complete water-quality dataset, containing 5,433 observations across 21 monitoring stations. Station identity is not used to filter these descriptive statistics.
- **Ramamangalam Monitoring Station:** A specific analysis for the Central Water Commission (CWC) monitoring station located in Ernakulam district, Kerala, on the Muvattupuzha River basin. This station has 359 recorded observations spanning 1980–2017. Non-water-quality identifier columns (serial number, administrative codes, latitude/longitude) were excluded from this analysis, as they are not water quality parameters.

1.2 Descriptive Statistics

Part A: Full Kerala CWC Dataset (All Monitoring Stations Combined)

For the aggregated dataset, missing or blank values were excluded parameter-by-parameter and never treated as zero, causing the valid sample size (N) to differ substantially across parameters. Mean, median, mode, minimum, maximum, range, sample standard deviation ($N-1$ denominator), sample variance ($N-1$ denominator), Q1, and Q3 (25th and 75th percentiles via linear interpolation) were calculated using only available numerical readings. Where multiple values shared the highest frequency, all tied modes are reported.

Descriptive Statistics Table – Well-Populated Parameters ($N \geq 150$)

The following parameters have at least 150 valid observations in the full dataset.

Table 1.1: Descriptive statistics – well-populated parameters ($N \geq 150$), All Stations Combined.

Parameter	**N**	**Mean**	**Median**	**Mode**	**Std Dev**	**Min**	**Max**	**Range**	**Variance**	**Q1**	**Q3**
Ammonia N (mgN/L)	455	0.1169	0.0720	0.05	0.1182	0.0000	0.7310	0.7310	0.0140	0.0500	0.1630
Carbonate (mg/L)	4841	0.7162	0.0000	0	2.5388	0.0000	48.0000	48.0000	6.4457	0.0000	0.0000
Calcium (mg/L)	5317	7.4847	4.8000	4.8	7.4196	0.0400	62.4000	62.3600	55.0508	4.0000	7.2200
Chloride (mg/L)	5305	12.9243	11.0000	9	7.0420	1.4400	97.6000	96.1600	49.5896	9.0000	15.3000
Dissolved oxygen (mg/L)	3693	7.0429	7.1000	7.2	1.0138	2.0000	27.0000	25.0000	1.0278	6.6000	7.6300
Total Dissolved Solids (mg/L)	1898	56.6396	38.0000	32	57.3045	10.0000	530.0000	520.0000	3283.8055	30.0000	56.0000
Bicarbonate (mg/L)	5187	26.6383	15.9800	12.2	30.0782	1.0250	281.8460	280.8210	904.7004	9.8000	29.0000
Magnesium (mg/L)	5317	4.2202	2.4300	1.458	5.6225	0.0000	83.0000	83.0000	31.6120	1.4580	4.3740
Sodium (mg/L)	4879	6.0278	4.0000	3	5.7328	1.0000	54.1000	53.1000	32.8655	3.0000	6.0000
Nitrite N+Nitrate N (mgN/L)	958	0.7652	0.3795	0.03	1.1808	0.0000	14.0060	14.0060	1.3942	0.1800	0.9000
Potential of Hydrogen (pH)	5212	7.1238	7.1200	6.9	0.4900	4.3200	10.0000	5.6800	0.2401	6.8000	7.4400
Sulphate (mg/L)	4066	2.5681	1.0000	1	4.5162	0.0000	72.0000	72.0000	20.3960	1.0000	2.0000
Total Phosphorus (mgP/L)	2801	0.1074	0.0660	0.01	0.7342	0.0000	37.0000	37.0000	0.5391	0.0200	0.1150
Arsenic (mg/L)	215	3.1535	2.0700	0.01	3.3221	0.0000	14.8000	14.8000	11.0364	0.6100	4.5450
Total Alkalinity (mg/L as $CaCO_3$)	4950	28.7566	17.0000	14	32.4278	0.8400	282.7000	281.8600	1051.5590	12.0000	29.0000
Cadmium (mg/L)	239	1.3213	0.0900	0	3.8059	0.0000	21.0000	21.0000	14.4850	0.0300	0.3300
Chromium (mg/L)	255	2.6846	1.1300	2	7.9706	0.0031	114.0400	114.0369	63.5300	0.5500	2.4450
Iron (mg/L)	618	0.2022	0.0700	0.1	0.8808	0.0000	20.0000	20.0000	0.7757	0.0300	0.1160
Total Hardness (mg $CaCO_3$/L)	5316	36.1711	22.0833	20	37.1565	3.0000	405.3333	402.3333	1380.6037	16.2500	36.2500
Nitrate N (mgN/L)	920	0.7557	0.3485	0.126	1.1888	0.0000	14.0000	14.0000	1.4132	0.1670	0.8815
Nickel (mg/L)	230	4.2177	2.9300	0	4.6801	0.0000	51.2400	51.2400	21.9031	1.4250	5.8750
Lead (mg/L)	258	2.7447	1.6240	1	4.1225	0.0000	41.4770	41.4770	16.9952	0.3085	3.7775
Zinc (mg/L)	251	36.6643	9.5000	5	319.3109	0.0000	5061.0000	5061.0000	101959.4339	3.0000	20.1100
Sodium Adsorption Ratio (%)	4815	0.4442	0.3853	0.3078	0.2494	0.0000	3.2673	3.2673	0.0622	0.2887	0.5348
Copper (mg/L)	260	5.5332	2.7200	1	11.3976	0.0000	130.7400	130.7400	129.9048	1.1575	5.5300

Descriptive Statistics Table – Sparse Parameters ($0 < N < 150$)

These parameters have fewer than 150 valid observations, yet their descriptive statistics are included because available values provide insight and should not be discarded.

Table 1.2: Descriptive statistics – sparse parameters ($0 < N < 150$), All Stations Combined.

Parameter	**N**	**Mean**	**Median**	**Mode**	**Std Dev**	**Min**	**Max**	**Range**	**Variance**	**Q1**	**Q3**
Hardness Calcium (mg $CaCO_3$/L)	60	51.7833	20.5000	12, 17	49.5784	7.0000	204.1680	197.1680	2458.0138	12.7500	88.0207
Mercury (mg/L)	46	0.1105	0.0400	0	0.1342	0.0000	0.5300	0.5300	0.0180	0.0000	0.2275

Notes and Observations (All Stations Combined)

Interpretation: The descriptive statistics show substantial variation among the water-quality parameters in the combined Kerala dataset. Parameters such as calcium, chloride, bicarbonate and total hardness have relatively large ranges, while pH shows comparatively limited variation. Several parameters have substantially fewer valid observations than others because of missing measurements. Therefore, the number of valid observations (N) should be considered when interpreting the descriptive statistics.

Note on Extreme Values: Some parameters exhibit large maximum values relative to their median values, indicating substantial dispersion and the possibility of extreme observations. These values are retained in the descriptive analysis because they are valid recorded measurements; detailed outlier analysis is addressed separately in Section 1.4.

- The following numerical parameters have no valid observations in the supplied dataset: Boron, Fluoride, Potassium, Hardness Magnesium, Manganese, and Nitrate.
- The combined dataset provides a larger sample than a single-station analysis, making the overall descriptive statistics more representative of the Kerala CWC dataset as a whole.
- Differences in sample size arise because parameters were not measured or recorded consistently for every observation, which requires missing readings to be excluded rather than imputed as zero.

Part B: Ramamangalam Monitoring Station

Of the 33 numerical water quality parameters in the source dataset, 26 have at least one valid reading at the Ramamangalam station and are summarized below. The remaining 7 (Boron, Fluoride, Potassium, Hardness Calcium, Hardness Magnesium, Manganese, and Nitrate) have no recorded values for Ramamangalam and are therefore excluded.

Missing or blank readings were excluded from every calculation (not treated as zero), ensuring statistics reflect only valid recorded values for each parameter, with sample sizes (N) varying accordingly. Mean, median, sample standard deviation, and sample variance were computed conventionally.

Descriptive Statistics Table – Well-Populated Parameters ($N \geq 150$)

These 14 parameters have sufficient valid readings (at least 150 of 359 records) for the statistics below to be considered reliable.

Table 1.3: Descriptive statistics – well-populated parameters ($N \geq 150$), Ramamangalam station.

Parameter	**N**	**Mean**	**Median**	**Mode**	**Std Dev**	**Min**	**Max**	**Range**	**Variance**	**Q1**	**Q3**
Carbonate (mg/L)	323	0.9165	0	0.0	1.9766	0	9.5374	9.5374	3.9068	0	0.5891
Calcium (mg/L)	351	4.6339	4.8	4.8	2.8939	0.04	28.19	28.15	8.3748	3.2	5.6
Chloride (mg/L)	351	11.3837	10	11.0	4.5083	2.87	42.7	39.83	20.3252	8.79	13
Dissolved oxygen (mg/L)	224	7.6196	7.6	7.0; 7.2	1.5028	3.7	27	23.3	2.2584	7.02	8.0175
Total Dissolved Solids (mg/L)	154	33.8442	33	28.0; 40.0	9.3145	11	75	64	86.7599	28	40
Bicarbonate (mg/L)	340	17.3133	14.64	8.5	11.7936	4.9	84.51	79.61	139.0893	8.5	19.9776
Magnesium (mg/L)	351	2.619	1.944	1.458	2.3967	0	19.683	19.683	5.744	1.458	2.916
Sodium (mg/L)	311	3.6649	3.4	3.0	1.8843	1	15.52	14.52	3.5505	2.8	4
Potential of Hydrogen (pH)	342	7.1057	7.115	6.9; 7.4	0.4148	6	8.43	2.43	0.1721	6.8225	7.4
Sulphate (mg/L)	237	1.2381	1	1.0	0.5974	0.5	5.6	5.1	0.3569	1	1.2
Total Phosphorus (mgP/L)	160	0.0849	0.07	0.01	0.0807	0.009	0.426	0.417	0.0065	0.02	0.11
Total Alkalinity (mg/L as $CaCO_3$)	333	17.9608	14	14.0	10.1932	5	76.3	71.3	103.9012	12	20
Total Hardness (mg $CaCO_3$/L)	351	22.3901	20	20.0	13.6482	5	110.0583	105.0583	186.2723	16	24
Sodium Adsorption Ratio (%)	311	0.3615	0.3488	0.3077793	0.1658	0.0926	1.527	1.4344	0.0275	0.2632	0.4149

Descriptive Statistics Table – Sparse Parameters ($N < 150$)

These 12 parameters have limited valid readings (mostly trace metals and nutrients sampled less frequently). Statistics are reported for completeness but should be interpreted with caution given the small sample sizes – several have fewer than 15 valid readings.

Table 1.4: Descriptive statistics – sparse parameters ($N < 150$), Ramamangalam station.

Parameter	N	Mean	Median	Mode	Std Dev	Min	Max	Range	Variance	Q1	Q3
Ammonia N (mgN/L)	26	0.1052	0.0785	0.05	0.0983	0.002	0.342	0.34	0.0097	0.05	0.1263
Nitrite N + Nitrate N (mgN/L)	52	0.6926	0.429	0.22; 0.78	0.6066	0.12	2.969	2.849	0.368	0.215	1.0422
Arsenic (mg/L)	12	3.1115	1.8425	0.01	3.2331	0.01	9.69	9.68	10.4528	0.76	4.7275
Cadmium (mg/L)	13	1.6474	0.136	0.1; 0.55	4.3855	0	16	16	19.2325	0.04	0.55
Chromium (mg/L)	14	1.3563	0.934	0.49	1.0335	0.22	3.24	3.02	1.0682	0.49	2.3075
Iron (mg/L)	36	0.7286	0.0625	0.032; 0.04; 0.09	3.317	0.009	20	19.991	11.0027	0.032	0.1555
Mercury (mg/L)	4	0.098	0.081	0.0; 0.032; 0.13; 0.23	0.1039	0	0.23	0.23	0.0108	0.024	0.155
Nitrate N (mgN/L)	48	0.6209	0.418	0.226; 0.276; 0.78	0.5143	0.101	2.116	2.015	0.2645	0.225	0.9685
Nickel (mg/L)	13	3.6248	2.24	1.85; 5.21	2.5284	0.038	8.1	8.062	6.3928	1.85	5.21
Lead (mg/L)	15	2.0563	1.634	0.21; 0.25; 3.91	1.7489	0.21	5.84	5.63	3.0586	0.265	3.2405
Zinc (mg/L)	13	13.9295	10.5	10.5	16.7003	0.06	60	59.94	278.8995	3.703	19.6
Copper (mg/L)	15	6.8187	2.9	2.98; 5.53	16.0826	0.4	64.57	64.17	258.6485	1.0705	4.265

Notes and Observations (Ramamangalam)

- Carbonate (mg/L) has a median of 0 – over half of all readings are exactly zero.
- Several sparse parameters (Arsenic, Cadmium, Copper, Zinc) show standard deviations larger than their means, worth flagging for outlier review.
- Dissolved Oxygen's maximum (27.0 mg/L) is far above its median and mean (both 7.6 mg/L) – a single high reading, not a shift in the overall distribution.

1.3 Outlier Detection (IQR Method / Z-Score)

Part A: Full Kerala CWC Dataset (All Monitoring Stations Combined)

For consistency and reliability, outlier analysis was restricted to the eleven parameters selected by the group based on sufficient data coverage (under ∼15% missing values). Parameters with more than 80% missing data (e.g., heavy metals, ammonia) were excluded from this analysis because their small effective sample sizes make outlier statistics unreliable. Two methods were compared: the IQR method (values beyond $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$) and the Z-score method ($|z| > 3$, i.e., more than 3 standard deviations from the mean).

Table 1.5: Outlier Detection Comparison – All Stations Combined

Parameter	n	Q1	Q3	IQR	IQR Outliers	Z-score Outliers	Skewness
pH	5212	6.80	7.44	0.64	62	32	-0.18
Chloride (mg/L)	5305	9.00	15.30	6.30	341	84	2.58
Sulphate (mg/L)	4066	1.00	2.00	1.00	662	67	6.91
Sodium (mg/L)	4879	3.00	6.00	3.00	643	137	2.97
Calcium (mg/L)	5317	4.00	7.22	3.22	727	169	2.71
Magnesium (mg/L)	5317	1.46	4.37	2.92	667	85	5.52
Bicarbonate (mg/L)	5187	9.80	29.00	19.20	517	135	3.22
Carbonate (mg/L)	4841	0.00	0.00	0.00	953	108	7.59
Total Alkalinity (mg/L)	4950	12.00	29.00	17.00	587	156	3.03
Total Hardness (mg/L)	5316	16.25	36.25	20.00	653	133	3.26
SAR (%)	4815	0.29	0.53	0.25	237	70	2.58

Notes and Observations (All Stations):

- Carbonate has $Q1 = Q3 = 0$, indicating that at least 75% of the recorded values are zero. Consequently, the IQR is zero and every non-zero value is flagged by the IQR rule. This should be interpreted cautiously because the large outlier count may reflect the zero-inflated distribution rather than 953 independent anomalous events.

- Across nearly every parameter, the IQR method flags substantially more outliers than the Z-score method. This is expected because Z-score outlier detection assumes an approximately bell-shaped (normal) distribution, whereas all parameters except pH are strongly right-skewed.
- In a skewed distribution, the mean and standard deviation are themselves pulled toward the long tail, which makes the Z-score threshold less sensitive to genuine high-end extremes. IQR, being based on ranks (quartiles) rather than the mean/SD, is not distorted by skew and is therefore the more defensible primary method for this dataset.
- pH is the one parameter close to symmetric (skewness = -0.18), where the two methods are in closer agreement (62 vs. 32 outliers). The pH outliers include values as low as 4.32 and as high as 10.0, well outside the typical 6.5–8.5 range for natural surface water. These are plausible signals of localized acidic or alkaline discharge (e.g., industrial effluent, agricultural runoff, or algal bloom activity) rather than measurement noise, and are worth cross-referencing against station and date in the temporal analysis.

Part B: Ramamangalam Monitoring Station

IQR and Z-score outlier detection were recomputed using only Ramamangalam readings. Sample sizes per parameter range from 237 (Sulphate) to 351 (Chloride, Calcium, Magnesium, Total Hardness), which is still sufficient for meaningful quartile and moment-based statistics.

Table 1.6: Outlier Detection Comparison – Ramamangalam Station

Parameter	n	Q1	Q3	IQR	IQR Outliers	Z-score Outliers	Skewness
pH	342	6.82	7.40	0.58	1	1	-0.12
Chloride (mg/L)	351	8.79	13.00	4.21	13	4	2.65
Sulphate (mg/L)	237	1.00	1.20	0.20	47	4	3.83
Sodium (mg/L)	311	2.80	4.00	1.20	17	9	3.46
Calcium (mg/L)	351	3.20	5.60	2.40	19	3	2.29
Magnesium (mg/L)	351	1.46	2.92	1.46	31	9	3.20
Bicarbonate (mg/L)	340	8.50	19.98	11.48	24	6	2.26
Carbonate (mg/L)	323	0.00	0.59	0.59	63	12	2.40
Total Alkalinity (mg/L)	333	12.00	20.00	8.00	28	7	2.24
Total Hardness (mg/L)	351	16.00	24.00	8.00	36	9	2.61
SAR (%)	311	0.26	0.41	0.15	13	7	2.75

Notes and Observations (Ramamangalam):

- As with the pooled dataset, IQR flags more outliers than Z-score for every skewed parameter, and the two methods converge only for pH, which is the one near-symmetric variable.

- Outlier counts are naturally smaller in absolute terms than the pooled analysis simply because the sample is smaller, but the proportional pattern (heavier skew leads to a larger IQR/Z-score gap) holds.

1.4 Correlation Matrix and Strongest Correlations

Part A: Full Kerala CWC Dataset (All Monitoring Stations Combined)

Pearson correlation coefficients were computed between all eleven reliable parameters. The eight strongest pairwise correlations are summarized below.

Table 1.7: Strongest Pairwise Correlations – All Stations Combined

Parameter 1	**Parameter 2**	**Correlation (r)**
Bicarbonate	Total Alkalinity	0.93
Magnesium	Total Hardness	0.91
Sodium	Total Alkalinity	0.86
Calcium	Total Hardness	0.86
Sodium	SAR	0.81
Sodium	Bicarbonate	0.76
Sodium	Calcium	0.75
Chloride	Sodium	0.72

Notes and Observations (All Stations):

- Several of the strongest correlations are partly structural rather than purely empirical: Total Hardness is calculated from Calcium and Magnesium, and Total Alkalinity is closely tied to Bicarbonate concentration. Their high correlations ($r = 0.86 - 0.93$) are expected by definition and confirm internal data consistency rather than revealing a new hydro-chemical relationship.

- Similarly, Sodium Adsorption Ratio is computed using sodium directly, explaining its strong link to Sodium ($r = 0.81$).

- The more genuinely informative relationship is the broad correlation of Sodium with Total Alkalinity, Bicarbonate, and Calcium ($r = 0.75 - 0.86$), none of which are related by formula. This pattern may be consistent with a shared geochemical source: weathering and dissolution of carbonate rock formations releases sodium, calcium, bicarbonate, and alkalinity together into the water. Stations with higher rock-water interaction or longer residence time tend to show elevated levels across all four parameters simultaneously.

- The moderate Chloride-Sodium correlation ($r = 0.72$) may additionally reflect anthropogenic inputs such as agricultural runoff or domestic wastewater, since chloride is not part of the natural carbonate-weathering pathway.

Part B: Ramamangalam Monitoring Station

Table 1.8: Strongest Pairwise Correlations – Ramamangalam Station

Parameter 1	Parameter 2	Correlation (r)
Bicarbonate	Total Alkalinity	0.93
Sodium	SAR	0.87
Magnesium	Total Hardness	0.86
Calcium	Total Hardness	0.71
Sulphate	Carbonate	0.50
Calcium	Carbonate	-0.46
Calcium	Total Alkalinity	-0.44
Bicarbonate	Carbonate	0.42

Notes and Observations (Ramamangalam):

- The Bicarbonate-Total Alkalinity and Magnesium/Calcium-Total Hardness correlations remain strong here, consistent with the pooled dataset (these are structural relationships).
- The notable difference is that Calcium shows a moderate negative correlation with Carbonate ($r = -0.46$) and Total Alkalinity ($r = -0.44$) at this station specifically, whereas the pooled (all-station) analysis showed these as positive.
- This divergence suggests Ramamangalam's local water chemistry, or a temporal shift specific to this station, does not follow the general basin-wide carbonate-weathering pattern discussed in the pooled analysis, and may be worth cross-referencing with the temporal/line-plot analysis for this station.

1.5 Skewness and Kurtosis

Part A: Full Kerala CWC Dataset (All Monitoring Stations Combined)

Skewness measures asymmetry (0 = symmetric, positive = long right tail) and kurtosis (reported here as excess kurtosis, where 0 = normal) measures tail heaviness relative to a normal distribution.

Notes and Observations (All Stations):

Table 1.9: Skewness and Kurtosis – All Stations Combined

Parameter	Skewness	Excess Kurtosis	Shape	Tails
pH	-0.18	1.00	Approx. symmetric	Heavy-tailed
Chloride (mg/L)	2.58	14.14	Right-skewed	Heavy-tailed
Sulphate (mg/L)	6.91	68.79	Right-skewed	Heavy-tailed
Sodium (mg/L)	2.97	11.32	Right-skewed	Heavy-tailed
Calcium (mg/L)	2.71	8.26	Right-skewed	Heavy-tailed
Magnesium (mg/L)	5.52	51.78	Right-skewed	Heavy-tailed
Bicarbonate (mg/L)	3.22	13.10	Right-skewed	Heavy-tailed
Carbonate (mg/L)	7.59	93.51	Right-skewed	Heavy-tailed
Total Alkalinity (mg/L)	3.03	10.63	Right-skewed	Heavy-tailed
Total Hardness (mg/L)	3.26	15.46	Right-skewed	Heavy-tailed
SAR (%)	2.58	14.60	Right-skewed	Heavy-tailed

- Every parameter except pH is strongly right-skewed with heavy tails (excess kurtosis well above zero, most extremely for Carbonate at 93.5 and Sulphate at 68.8).
- Practically, this means the large majority of readings across stations cluster at relatively low, similar concentrations, while a small number of readings spike far above the typical range. This pattern indicates that most observations are concentrated at relatively low values, with a smaller number of substantially higher observations. Such extreme values may reflect localized hydrological or anthropogenic influences, but their specific causes require comparison with station, date, and contextual information.
- pH stands apart as the only approximately symmetric, approximately bell-shaped parameter (skewness = -0.18). This matters methodologically: pH is approximately symmetric and therefore is more compatible with normal-based interpretations than the strongly right-skewed parameters.

Part B: Ramamangalam Monitoring Station

Notes and Observations (Ramamangalam):

- The distribution shapes at Ramamangalam mirror the pooled dataset: every parameter except pH is strongly right-skewed with heavy tails.
- pH has the most approximately symmetric distribution among the parameters examined (skewness = -0.12), here also showing near-normal (slightly negative) excess kurtosis rather than the heavy tails seen pooled.

Table 1.10: Skewness and Kurtosis – Ramamangalam Station

Parameter	**Skewness**	**Excess Kurtosis**	**Shape**	**Tails**
pH	-0.12	-0.15	Approx. symmetric	Near-normal tails
Chloride (mg/L)	2.65	12.50	Right-skewed	Heavy-tailed
Sulphate (mg/L)	3.83	20.46	Right-skewed	Heavy-tailed
Sodium (mg/L)	3.46	16.54	Right-skewed	Heavy-tailed
Calcium (mg/L)	2.29	13.31	Right-skewed	Heavy-tailed
Magnesium (mg/L)	3.20	13.04	Right-skewed	Heavy-tailed
Bicarbonate (mg/L)	2.26	7.07	Right-skewed	Heavy-tailed
Carbonate (mg/L)	2.40	5.00	Right-skewed	Heavy-tailed
Total Alkalinity (mg/L)	2.24	6.10	Right-skewed	Heavy-tailed
Total Hardness (mg/L)	2.61	9.23	Right-skewed	Heavy-tailed
SAR (%)	2.75	12.94	Right-skewed	Heavy-tailed

Chapter 2

Visualization

2.1 Histograms

Pre-Analysis Data Preparation

- Blank Data was considered None instead of zero, so they were not included in any calculations.
- Blank readings kept as missing and excluded from every statistic — never filled with 0, because 0 mg/L is a real measurement for some parameters. This is why sample size N varies by parameter.

Parameter Selection: Those parameters were selected that have $N \geq 150$. So only 14 parameters were analyzed in this document. Because several parameters have extremely sparse observations, the visualizations are restricted to parameters and stations with sufficient data coverage to produce reliable distributions and comparisons; descriptive statistics for parameters with available observations are nevertheless reported.

Station Selection: A station is included only if every one of the 14 parameters has at least 100 valid readings there. This is stricter than ranking by record count — a station can have many records but be missing an entire parameter, leaving holes in the comparison grid. Exactly **8 stations** clear the rule, and they cover 8 different rivers across 7 districts.

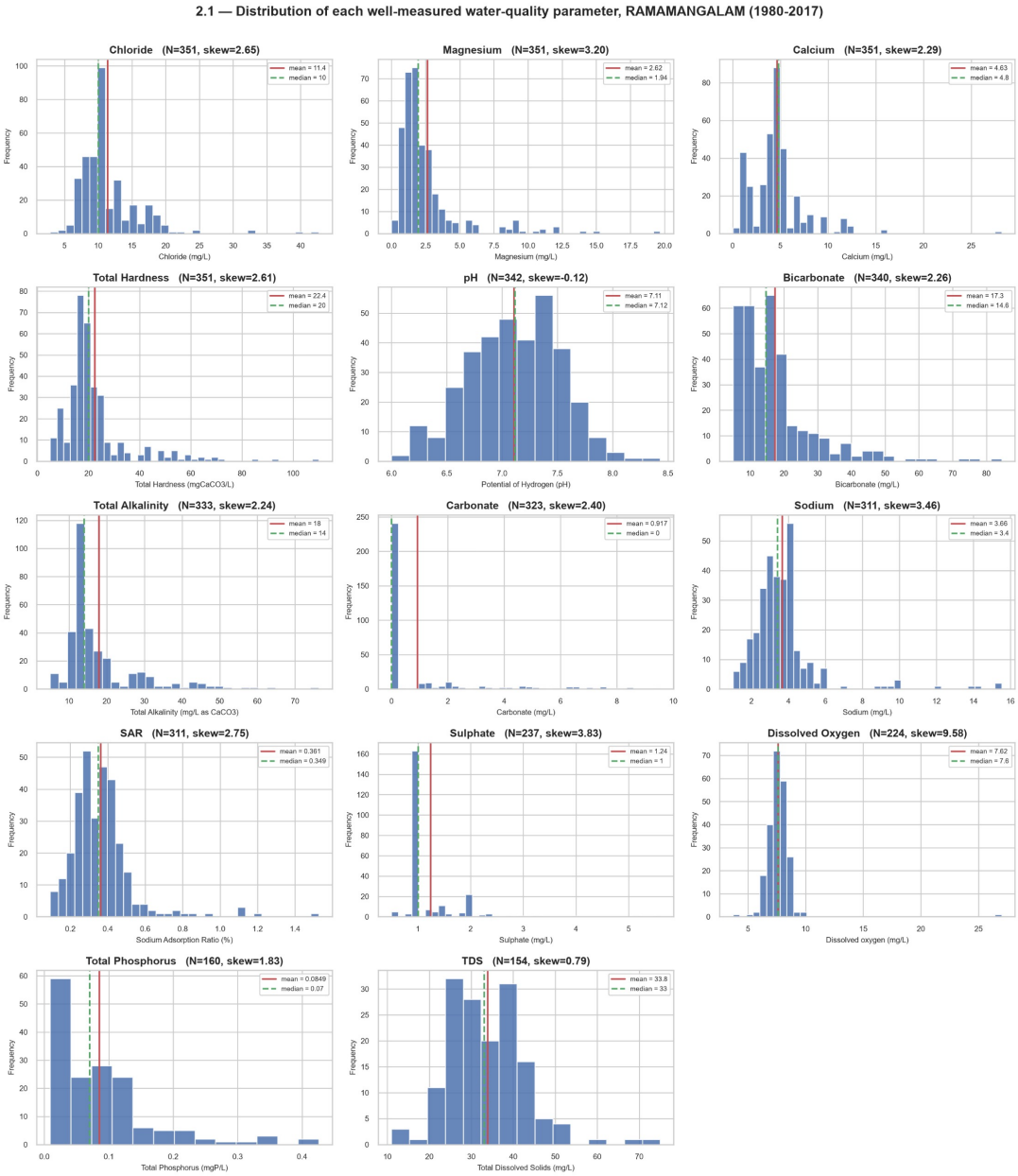


Figure 2.1: Distribution of each well-measured water-quality parameter, Ramamangalam (1980–2017).

Procedure followed: After the 14 parameters were selected, histograms were plotted. Bin widths use the **Freedman–Diaconis rule** (scales with IQR, so a single extreme reading can't flatten the plot). Each panel shows the mean (solid red) and median (dashed green) — their gap is a visual read of skew.

Abstract Inferences: Across the eight well-sampled stations, 91.9% of valid total-hardness observations fall below 60 mg $CaCO_3$/L and 99.8% fall below 200 mg $CaCO_3$/L. At Ramamangalam specifically, the median total hardness is 20 mg $CaCO_3$/L, indicating predominantly soft water — an order of magnitude below what a limestone catchment would yield.

This is consistent with the geological characteristics of the region, although local

hydrochemical conditions and sampling variability may also contribute. These rivers drain **crystalline hard-rock terrain — gneiss and charnockite** — which weathers slowly and releases very little carbonate. There is no substantial mineral source for the water to pick up, and the state's very high rainfall continually flushes what little there is. Soft, dilute, near-neutral water is the expected end product, and that is exactly what the histograms show.

Main findings:

- **13 of 14 parameters are right-skewed, 12 strongly.** Expected for solute concentrations in a monsoon-fed river: a floor set by geology, no ceiling. So median and IQR — not mean and SD — are the correct summaries.
- **pH is the only symmetric parameter.** Chemically expected: pH is logarithmic, and the river is bicarbonate-buffered.
- **Total Phosphorus is the most genuinely variable parameter** (CV 0.95, spanning two orders of magnitude) — a marker of episodic input rather than steady background.
- **Calcium is the one right-skewed parameter whose mean sits *left* of its median.** It has a second cluster of low readings, and 67 of the 69 readings ≤ 2.0 mg/L are from the 1980s.
- **For irrigation, the verdict is unambiguous.** The Sodium Adsorption Ratio never exceeds **3.27** across all 2,600+ readings, against a threshold of 10 for "excellent". 100% of samples fall in the excellent category, with the highest reading barely a third of the way to the first boundary. Sodium is not accumulating in these waters at levels that would threaten soil structure.

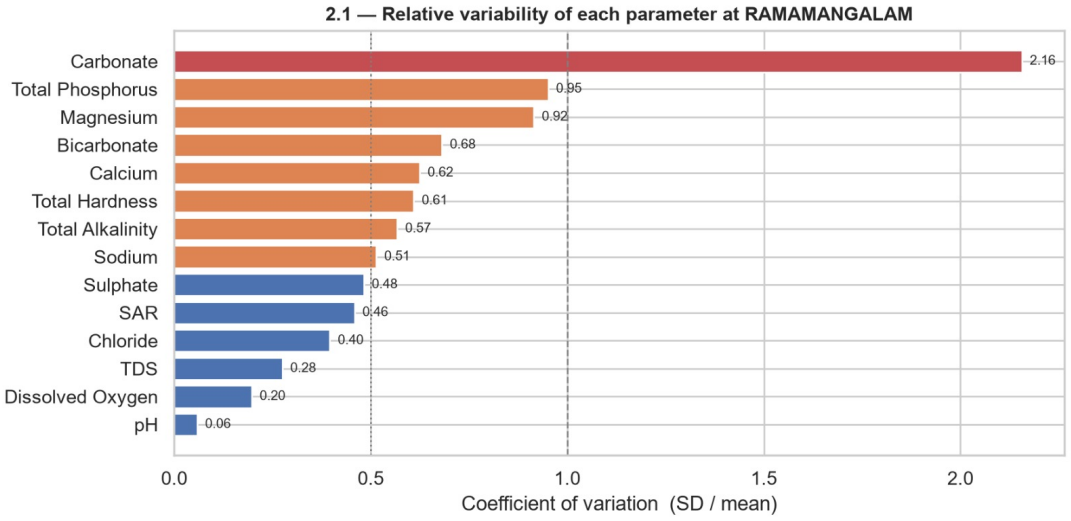


Figure 2.2: Relative variability of each parameter at Ramamangalam (1980–2017).

Table 2.1: Parameter Variability Metrics – Ramamangalam

Parameter	**Mean**	**Median**	**Std dev**	**CV**	**Min**	**Max**	**Range**
Carbonate	0.917	0.000	1.977	2.157	0.000	9.537	9.537
Total Phosphorus	0.085	0.070	0.081	0.951	0.009	0.426	0.417
Magnesium	2.619	1.944	2.397	0.915	0.000	19.683	19.683
Bicarbonate	17.313	14.640	11.794	0.681	4.900	84.510	79.610
Calcium	4.634	4.800	2.894	0.625	0.040	28.190	28.150
Total Hardness	22.390	20.000	13.648	0.610	5.000	110.058	105.058
Total Alkalinity	17.961	14.000	10.193	0.568	5.000	76.300	71.300
Sodium	3.665	3.400	1.884	0.514	1.000	15.520	14.520
Sulphate	1.238	1.000	0.597	0.482	0.500	5.600	5.100
SAR	0.361	0.349	0.166	0.459	0.093	1.527	1.434
Chloride	11.384	10.000	4.508	0.396	2.870	42.700	39.830
TDS	33.844	33.000	9.314	0.275	11.000	75.000	64.000
Dissolved Oxygen	7.620	7.600	1.503	0.197	3.700	27.000	23.300
pH	7.106	7.115	0.415	0.058	6.000	8.430	2.430

2.2 Boxplots (Station-wise Analysis)

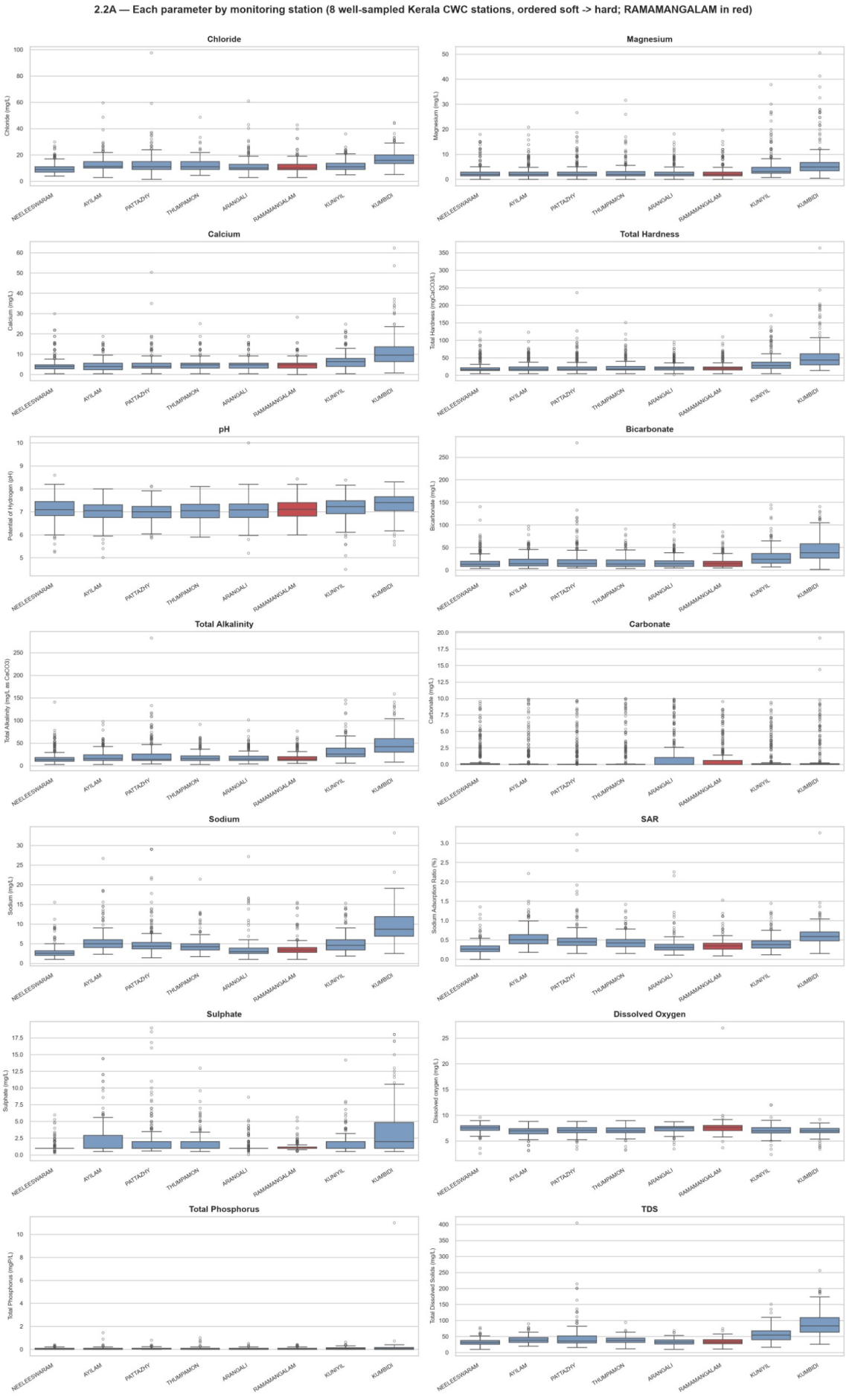


Figure 2.3: Each parameter by monitoring station (8 well-sampled Kerala CWC stations

Table 2.2: Pooled across all 8 stations

Parameter	N	Min	Q1	Median	Q3	Max	Range	IQR
Chloride	2,610	1.44	9.00	11.00	15.00	97.60	96.16	6.00
Magnesium	2,621	0.00	1.46	2.43	3.89	50.54	50.54	2.43
Calcium	2,621	0.04	3.20	4.80	6.40	62.40	62.36	3.20
Total Hardness	2,621	3.00	16.00	20.00	32.00	364.00	361.00	16.00
pH	2,554	4.50	6.80	7.11	7.40	10.00	5.50	0.60
Bicarbonate	2,545	1.90	9.80	17.00	28.98	281.85	279.95	19.18
Total Alkalinity	2,478	3.03	12.00	17.48	30.00	282.70	279.67	18.00
Carbonate	2,409	0.00	0.00	0.00	0.11	19.20	19.20	0.11
Sodium	2,316	1.00	3.00	4.00	5.80	33.20	32.20	2.80
SAR	2,307	0.00	0.29	0.39	0.52	3.27	3.27	0.23
Sulphate	1,717	0.08	1.00	1.00	2.00	19.00	18.92	1.00
Dissolved Oxygen	1,634	2.40	6.70	7.20	7.78	27.00	24.60	1.08
Total Phosphorus	1,224	0.01	0.02	0.07	0.12	11.00	10.99	0.10
TDS	1,087	10.00	30.00	38.00	54.00	405.00	395.00	24.00

Main Findings

One station accounts for nearly all the between-river variation. KUMBIDI has the highest median of the eight for 12 of the 14 parameters. This pattern may be related to geographic factors and is consistent across all of those parameters. KUMBIDI is the only station of the eight on the **Bharathapuzha in Palakkad** — the eastern rain-shadow district, where rainfall is markedly lower than on the western slopes. Less rain may lead to less dilution and stronger evaporative concentration, which is consistent with **every dissolved species rising together.** The two parameters that *don't* follow this pattern may further support the mechanism. **Dissolved oxygen at KUMBIDI is at the bottom of the range, not the top** — which is consistent with reduced flow and weaker re-aeration. And **the median carbonate concentration is zero at all eight stations**, although occasional non-zero observations occur.

Beyond KUMBIDI, the rivers are strikingly alike. KUNIYIL is a weaker second, and the remaining six cluster tightly — Total Hardness 16.67–20.0, pH 7.00–7.12, TDS 32–38.5. So these are not a smooth gradient of eight distinct rivers but one clearly distinct station, one intermediate, and a **homogeneous soft-water group** that includes Ramamangalam.

Two parameters do not distinguish the stations at all. Total Phosphorus ($p = 0.25$) and Carbonate ($p = 0.81$) are the only non-significant results, with all eight phosphorus medians falling in the narrow band 0.059–0.078 mg/L. This is worth comparing with 2.1, where Total Phosphorus was the most variable parameter at a single station (CV 0.95).

A parameter that varies enormously within each river but barely between them varies in time rather than space, which a station-wise summary alone would completely conceal.

Statistical significance and practical size diverge for pH and Dissolved Oxygen. Both separate the stations with overwhelming significance ($p < 0.0001$), yet in magnitude, they are trivial — pH spans just 7.00 to 7.40 across all eight rivers. With thousands of samples, a difference of 0.4 pH units is easily significant, so the effect size ($\eta^2 = 0.055$ for pH, 0.084 for Dissolved Oxygen) is the figure worth quoting, not the p-value.

Where Ramamangalam sits: Its medians are at or just below the middle of the eight for every mineral parameter (Total Hardness 20.0, Chloride 10.0, TDS 33.0), its IQRs are among the narrowest (Total Hardness 8.0, TDS 12.0, the joint-lowest of the eight), and it has the highest median Dissolved Oxygen of the group at 7.60 mg/L. Though NEELEESWARAM (7.58) and ARANGALI (7.54) are within 0.1, so it is better described as belonging to the best-oxygenated group than as a clear first. Combined with a near-neutral pH of 7.12, its distribution is that of a stable, soft, well-oxygenated river reach.

2.3 Correlation Heatmap

Ramamangalam Station is in the Ernakulam district, Kerala. It is a locality on the Muvattupuzha River basin.

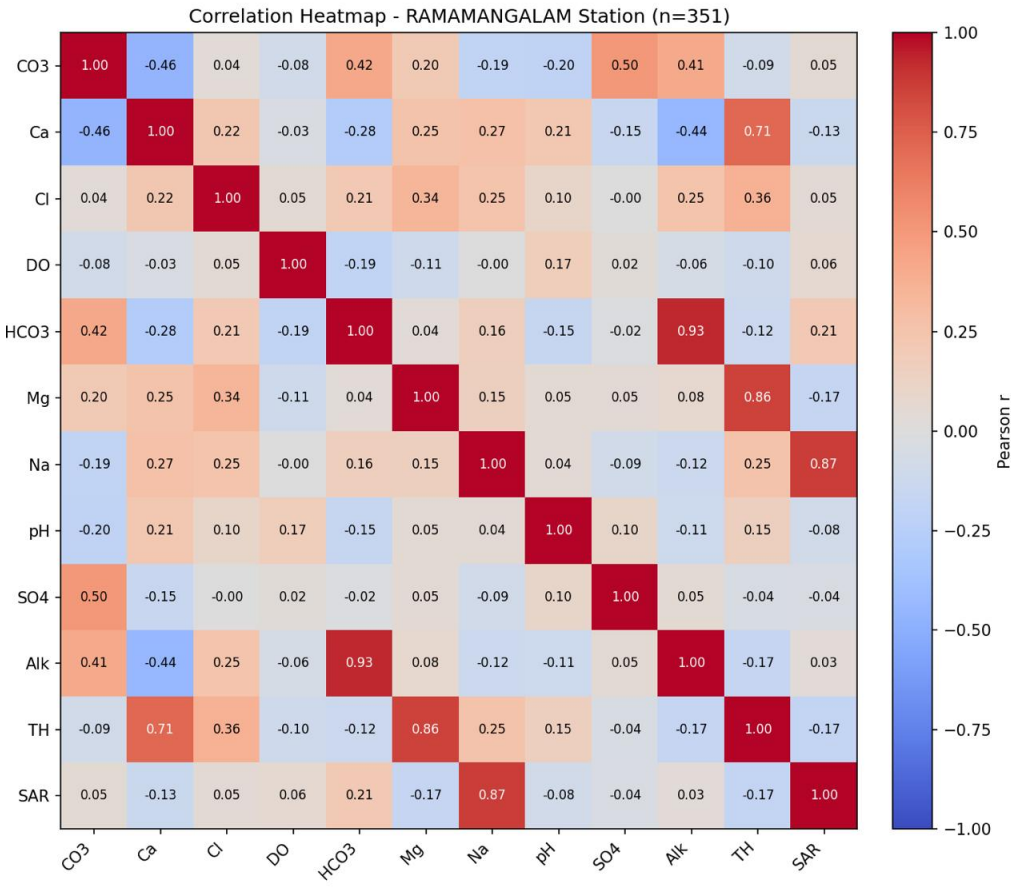


Figure 2.4: Correlation Heatmap - Ramamangalam Station ($n = 351$). Correlations are based on pairwise available observations; therefore, the effective sample size may vary between parameter pairs.

Strong Positive Correlations:

- **Bicarbonate-Alkalinity ($r = 0.93$):** Total alkalinity in these waters is almost entirely governed by bicarbonate ion concentration. This relationship occurs because bicarbonate (HCO_3^-) is typically the predominant contributor to total alkalinity, particularly within the pH range commonly encountered in natural waters. Total alkalinity represents the combined concentration of acid-neutralizing species, primarily bicarbonate and carbonate, with smaller contributions from hydroxide and other weak acid systems.

- **Sodium-SAR ($r = 0.87$):** A positive correlation between sodium concentration and Sodium Adsorption Ratio (SAR) is generally expected in water-quality data. SAR evaluates the relative abundance of sodium compared with calcium and magnesium, which are the principal competing cations in determining sodium hazard. Since sodium concentration is present directly in the numerator of the SAR calculation, an increase in sodium concentration generally results in an increase in SAR, provided that calcium and magnesium concentrations remain relatively comparable.

- **Magnesium-Total Hardness ($r = 0.86$):** A positive correlation between magnesium concentration and total hardness is generally observed in natural water sources because magnesium is one of the principal contributors to water hardness, along with calcium. Total hardness represents the combined concentration of calcium and magnesium ions, expressed as calcium carbonate ($CaCO_3$) equivalents.

- **Calcium-Total Hardness ($r = 0.71$):** Similar to magnesium, calcium is a major contributor to water hardness. An increase in calcium concentration generally results in an increase in total hardness, particularly when magnesium concentrations remain relatively stable.

Notable Negative Correlations:

- **Carbonate-Calcium ($r = -0.46$):** This negative relationship may be attributed to the interaction between calcium and carbonate ions in water. Calcium can react with carbonate to form relatively insoluble calcium carbonate ($CaCO_3$), which can reduce the concentration of dissolved carbonate.

- **Calcium-Alkalinity ($r = -0.44$):** This negative association may be influenced by the carbonate-bicarbonate equilibrium and the precipitation or dissolution of calcium carbonate. Higher calcium concentrations can promote the formation of calcium carbonate, potentially reducing the concentration of carbonate species contributing to alkalinity.

Weak Correlations:

- Dissolved Oxygen (DO) shows near-zero correlation with almost every parameter ($|r| \leq 0.19$ across the heatmap). This suggests DO at this station is controlled more by physical/biological factors (reaeration, temperature, photosynthesis-respiration cycles) than by the mineral/ionic composition of the water.

2.4 Line Plots (Temporal Analysis)

Potential of Hydrogen (pH)

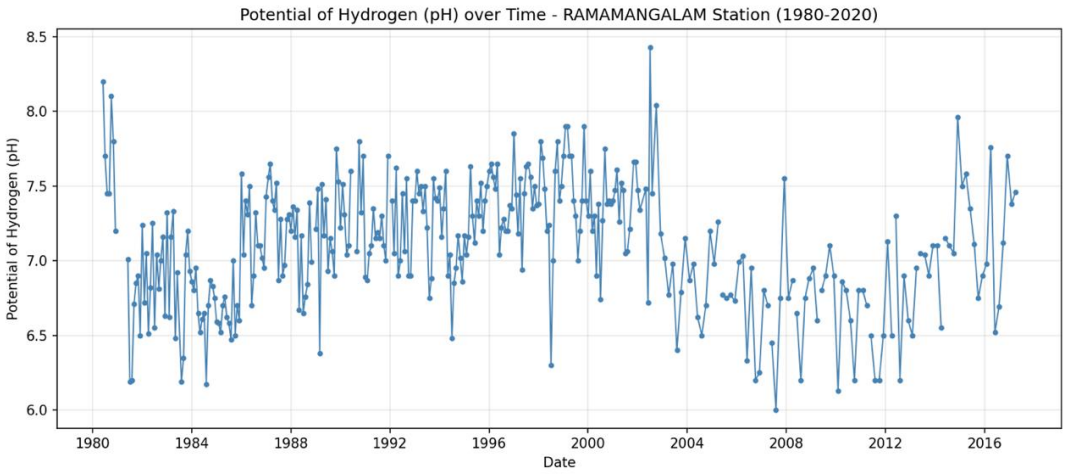


Figure 2.5: Potential of Hydrogen (pH) over Time - Ramamangalam Station (1980–2017).

The pH values show considerable short-term variability, generally remaining between about 6.5 and 7.7, with a few distinct excursions toward more acidic and alkaline conditions. The most prominent observations are the high values around 1980–81 and 2002, and the low values around 1982, 2007, and 2010–12.

The relatively neutral to slightly alkaline character of the water is consistent with the known chemistry of the Muvattupuzha basin. At Ramamangalam, bicarbonate is reported as an important anion, which indicates the influence of the carbonate-bicarbonate buffering system. This system tends to resist large changes in pH and helps explain why most observations cluster around neutral to mildly alkaline conditions. The pronounced fluctuations, however, suggest that this buffering capacity is periodically modified by seasonal hydrology and changes in the sources of water reaching the river.

Dissolved Oxygen (mg/L)

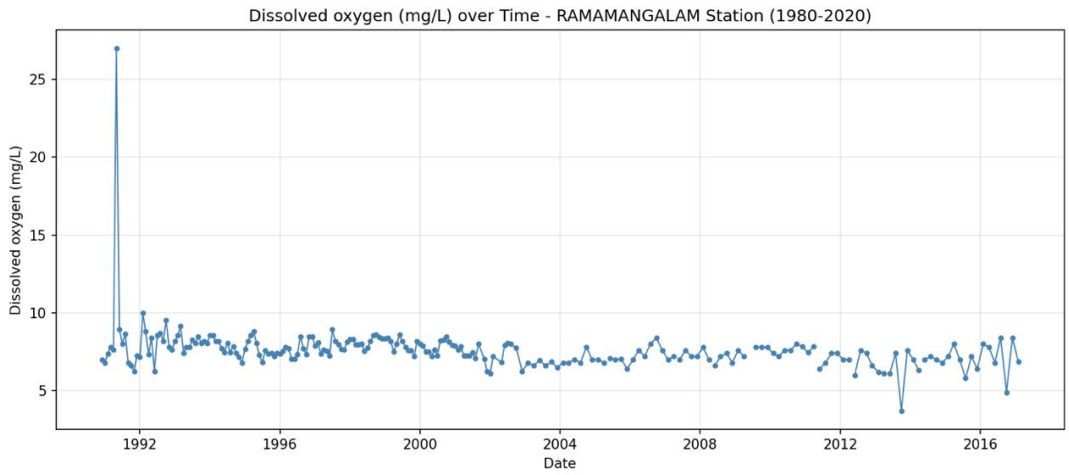


Figure 2.6: Dissolved oxygen (mg/L) over Time - Ramamangalam Station (1980–2017).

Dissolved oxygen remains mostly stable between 6−9 mg/L, indicating generally favourable oxygen conditions. A major anomaly is the exceptionally high value of ~ 27 mg/L around 1991, which is likely an outlier or may reflect sampling/measurement conditions rather than a persistent environmental condition. After the early 1990s, DO gradually shows a slight decline, with values becoming more frequently concentrated around $6 - 8$ mg/L particularly after 2002.

The relatively good DO levels suggest that the river generally retains adequate oxygenation, potentially related to flow and turbulence associated with the river system and monsoonal discharge. The occasional low values, particularly around 2013–2017, could indicate increased organic matter and wastewater inputs, as microbial decomposition consumes dissolved oxygen.

Chloride (mg/L)

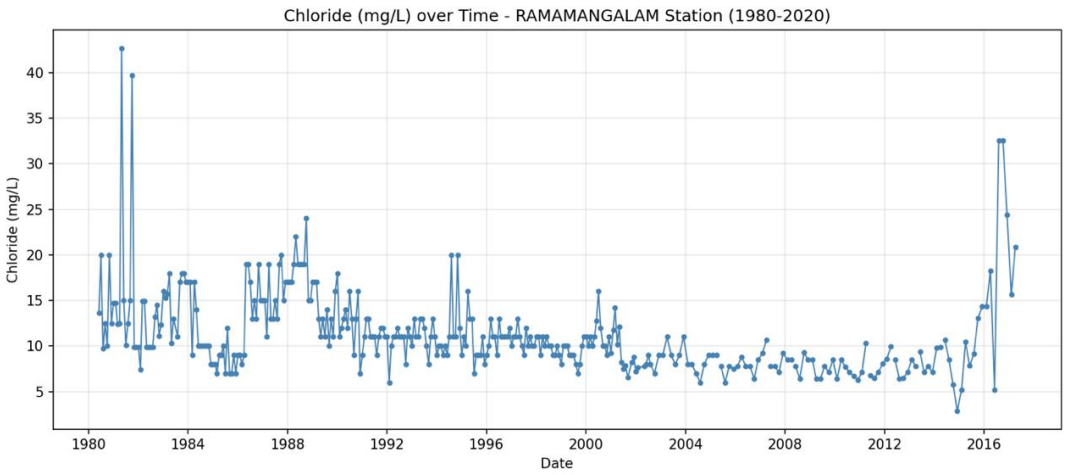


Figure 2.7: Chloride (mg/L) over Time - Ramamangalam Station (1980–2017).

Chloride concentrations show high variability during the 1980s, including an exceptional spike of about 42 mg/L around 1981–82. From the early 1990s onward, concentrations generally decline and remain mostly around $7 - 12$ mg/L, suggesting relatively stable chloride conditions. Another pronounced increase occurs around 2016–17, when chloride rises to approximately $30 - 33$ mg/L.

The relatively low chloride levels through most of the record indicate limited persistent salinity influence at Ramamangalam. The sharp spikes are more likely associated with episodic inputs or changes in river discharge rather than a continuous source. At Ramamangalam, monsoonal runoff can rapidly alter river chemistry through dilution or transport of dissolved ions from agricultural and settled areas. The late-period chloride increase could potentially indicate greater anthropogenic influence, such as domestic wastewater, agricultural return flow, or increased urban activity.

Total Hardness ($mgCaCO_3/L$)

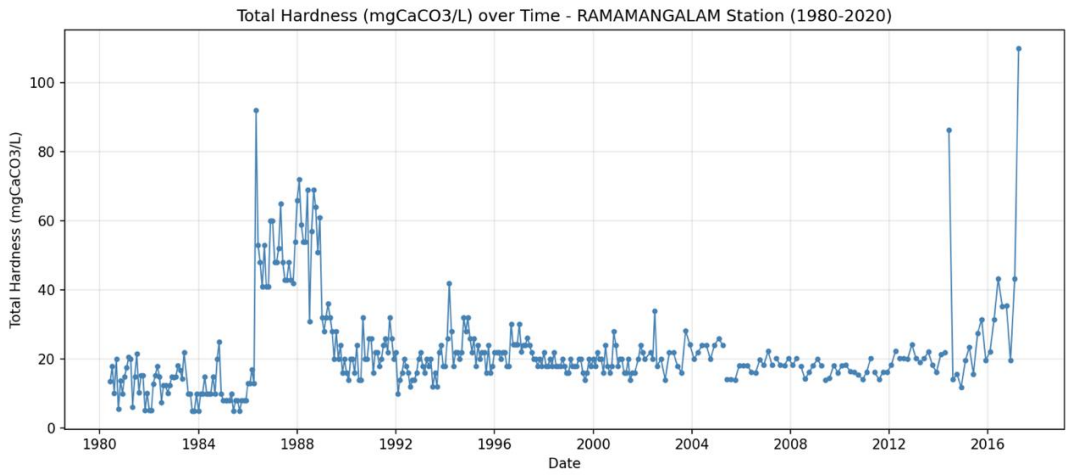


Figure 2.8: Total Hardness ($mgCaCO_3/L$) over Time - Ramamangalam Station (1980–2017).

Total hardness is highly variable in the 1980s, with a major rise around 1987–1989, reaching about 90 mg/L as $CaCO_3$, compared with mostly $5-25$ mg/L in the preceding years. After 1990, hardness generally decreases and remains relatively stable around $15-25$ mg/L through 2014. The later period shows renewed variability, including sharp peaks of approximately 86 mg/L (2014) and 110 mg/L (2017).

The high hardness episodes likely indicate periods of increased calcium and magnesium inputs, potentially from enhanced weathering/mineral dissolution and changes in river discharge. At Ramamangalam, variations in monsoonal flow can alter the balance between dilution and mineral-rich runoff, potentially producing substantial changes in hardness. The isolated high values in the later years could also reflect episodic runoff or localized anthropogenic inputs, but their sharp and isolated nature means they should also be treated as potential anomalous observations or sampling-related effects rather than evidence of a continuous increase in hardness.

2.5 Univariate Scatter Plots

Potential of Hydrogen (pH)

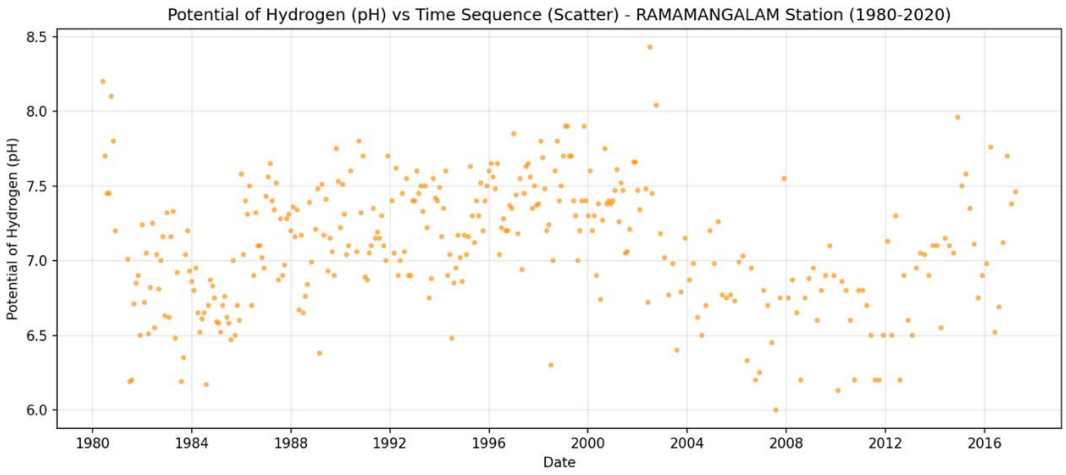


Figure 2.9: Potential of Hydrogen (pH) vs Time Sequence - Ramamangalam Station (1980–2017).

The scatter plot reveals distinct periods of clustering rather than a consistent long-term trend. Values are relatively low during 1982–1986, shift upward and cluster around 7.2–7.6 during 1987–2002, and then return to lower values, particularly during 2004–2012. The isolated values above 8 and below 6.5 represent unusual excursions that may reflect short-term changes in sampling conditions or water chemistry.

At Ramamangalam, this pattern can plausibly be linked to the strong seasonal hydrology of the Muvattupuzha basin. Monsoonal rainfall and runoff can dilute the river and introduce organic matter, potentially lowering pH, while periods of lower flow can increase the influence of carbonate/bicarbonate-rich groundwater and mineral weathering, maintaining slightly alkaline conditions.

Dissolved Oxygen (mg/L)

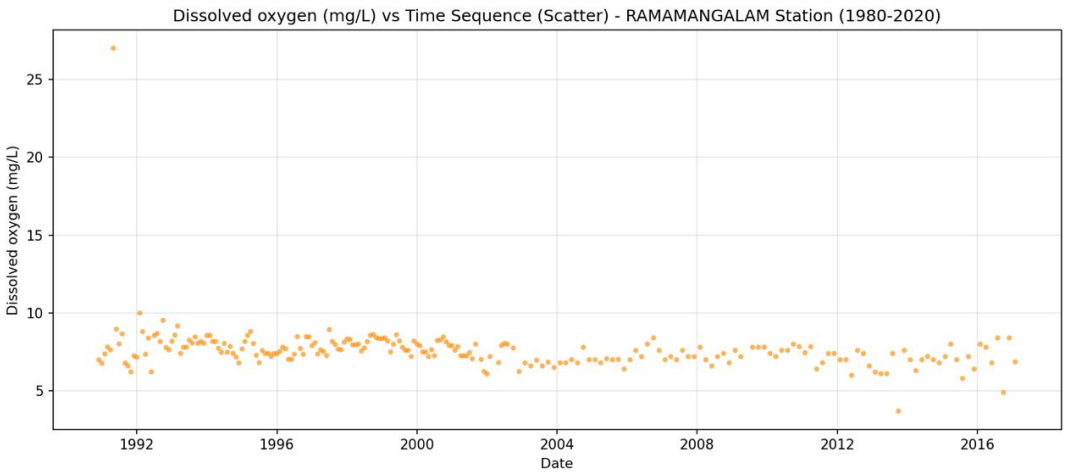


Figure 2.10: Dissolved oxygen (mg/L) vs Time Sequence - Ramamangalam Station (1980–2017).

DO is mostly clustered around $6 - 9$ mg/L, with a slight tendency toward lower values after the early 2000s. The extremely high value of ~ 27 mg/L around 1991 is a clear outlier and is more likely related to measurement/sampling conditions than the actual long-term river condition. A few low observations around 2013–2017 indicate episodic oxygen depletion.

The generally adequate DO suggests that river flow and natural reaeration at Ramamangalam help maintain oxygen levels, particularly during periods of stronger flow. The lower values in the later period could reflect increased organic matter and wastewater inputs from settlements and agricultural areas; microbial decomposition would consume oxygen. However, because the decline is not continuous and most values remain above ~ 6 mg/L, the data suggest episodic rather than persistent oxygen stress.

Chloride (mg/L)

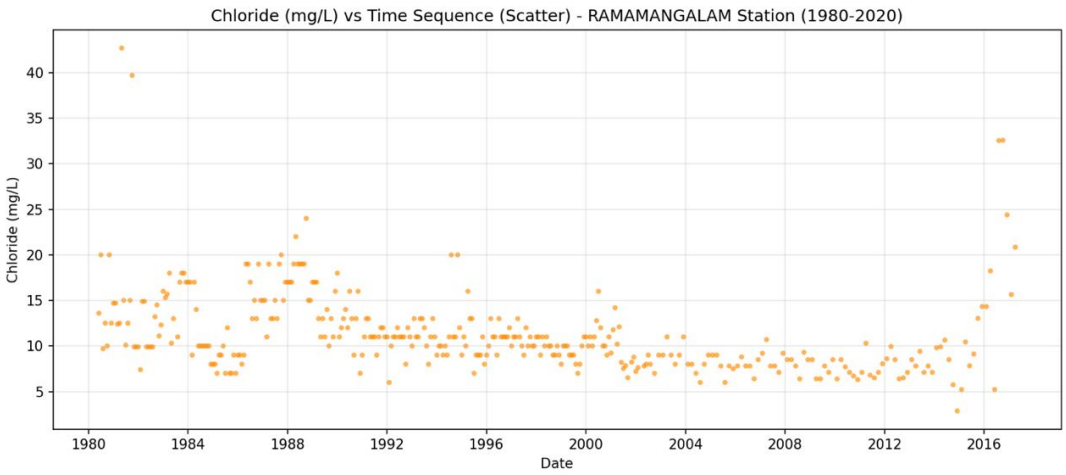


Figure 2.11: Chloride (mg/L) vs Time Sequence - Ramamangalam Station (1980–2017).

Chloride shows a clear shift from higher and more variable concentrations during the 1980s to lower, more stable values after $\sim$ 2000, with most later observations around $6 - 10$ mg/L. The major anomalies are the ~ 42 mg/L spike around 1981, several values around $20 - 24$ mg/L in the late 1980s/early 1990s, and a renewed cluster of high values ($\sim 25 - 33$ mg/L) around 2016–17.

The long-term reduction suggests that the river's chloride load was not persistently increasing, and may reflect changes in dilution, runoff, or sources of dissolved ions. The sharp late-period increase is more significant because it occurs after a long period of low concentrations and could indicate episodic domestic wastewater, agricultural runoff, or increased human activity in the basin. Since chloride is relatively conservative, such spikes can be useful indicators of changes in anthropogenic inputs or hydrological conditions.

Total Hardness ($mgCaCO_3/L$)

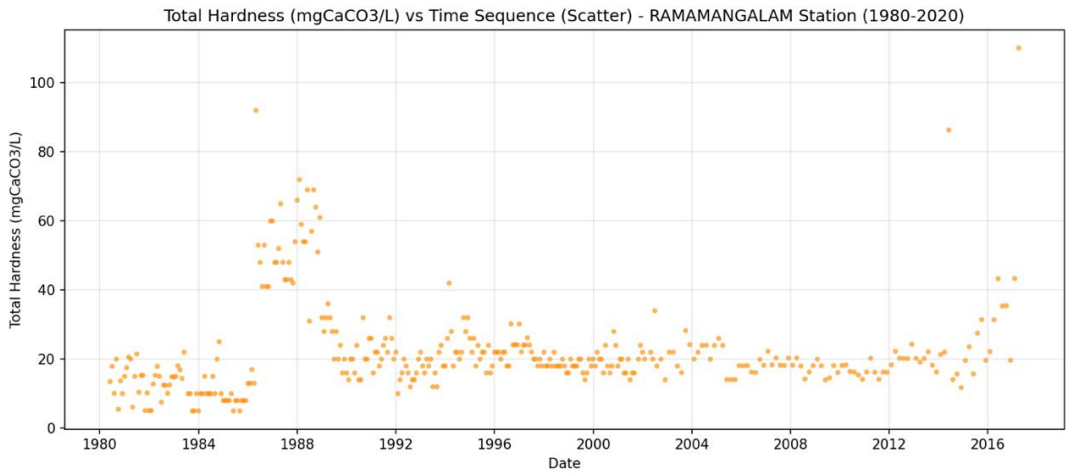


Figure 2.12: Total Hardness ($mgCaCO_3/L$) vs Time Sequence - Ramamangalam Station (1980–2017).

Total hardness shows a major shift in the late 1980s, rising from mostly $5 - 25$ mg/L to a concentrated cluster of $40 - 70$ mg/L around 1987–1989. After 1990, values largely return to $15 - 25$ mg/L and remain relatively stable until about 2014. The later period contains a few strong spikes, reaching ~ 86 mg/L in 2014 and ~ 110 mg/L in 2017.

The sharp 1987–89 increase suggests a temporary change in the calcium-magnesium inputs or hydrological conditions, potentially enhanced mineral weathering or reduced dilution during low-flow periods. The isolated high values after 2014 could reflect episodic mineral-rich runoff, groundwater contribution, or anthropogenic inputs from agricultural and developed areas around the Muvattupuzha basin. Since these peaks are isolated rather than sustained, they are better interpreted as episodic changes in water chemistry (or possible anomalous measurements) rather than evidence of a continuous deterioration in hardness.

References

Central Water Commission (CWC). (2020). *Surface Water Quality Dataset: Kerala (1961–2020).* [Data set].